

MCS-023 (DBMS)

1. Explain the Two-Phase Locking Protocol. How does it guarantee serializability?
2. What is deadlock and how can it be prevented? Describe common deadlock-prevention schemes.
3. Compare primary, secondary and clustering indexes. Which of these indexes are dense and which are not? How is implementation of clustering indexes performed?
4. How do B-tree indexes differ from binary search tree indexes?
5. What is a B⁺-tree? Why is a B⁺-tree better structure than a B-Tree for implementation of an index sequential file?
6. Write SQL commands for each of the following:
 - (i) Creation of views;
 - (ii) Creation of sequences;
 - (iii) Outer join;
 - (iv) To give access permission to any user.
7. Why is data replication useful in Distributed Database Management System (DDBMS)? Briefly discuss the terms 'Complete Replication' and 'Selective Replication'.
8. Explain the term data replication and data fragmentation with suitable example.
9. What are integrity constraints? Explain the various types of integrity constraints.
10. Describe the following client-server architectures with the help of a diagram:
 - (i) 2-tier
 - (ii) 3-tier
11. What is a hashed file organization and inverted file organisation? What are its advantages and disadvantages?
12. Explain the meaning of the following two keywords: Commit and Rollback. Write a code fragment for transferring money from account A to account B and show the uses of Commit and Rollback.
13. What are Checkpoints? How does this technique of Checkpoints contribute to database recovery? Give suitable example.
14. What is a data dictionary? What should be included in data dictionary?
15. What is Optimization in DBMS? How does relational algebra help in Optimizing the Query? Discuss with suitable example. List the operators used in relational algebra and briefly explain them.
16. Explain log-based recovery scheme and shadow paging recovery scheme with the help of an example.